

ML: TASK-04 Study of Classification Evaluation Metrics PDF Report

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Collab Notebook link :

https://colab.research.google.com/drive/1DhiydIUPKXIdYrn6_VCJp7feMkwwrH9#scrollTo=VLUvWL3HZvPb

Output screenshots :

About the dataset :

The titanic dataset was downloaded from Kaggle. It shows whether a passenger has survived or not based on different attributes.

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype  
---  -
 0   PassengerId   891 non-null    int64  
 1   Survived      891 non-null    int64  
 2   Pclass        891 non-null    int64  
 3   Name          891 non-null    object  
 4   Sex           891 non-null    object  
 5   Age           714 non-null    float64 
 6   SibSp         891 non-null    int64  
 7   Parch         891 non-null    int64  
 8   Ticket        891 non-null    object  
 9   Fare          891 non-null    float64 
10   Cabin         204 non-null    object  
11   Embarked      889 non-null    object  
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Classification Report :

```
#Classification Report  
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.80	0.84	0.82	105
1	0.75	0.70	0.73	74
accuracy			0.78	179
macro avg	0.78	0.77	0.77	179
weighted avg	0.78	0.78	0.78	179

Confusion matrix :

```
#Confusion Matrix  
cm = confusion_matrix(y_test, y_pred)
```

```
print(cm)
```

```
[[88 17]  
 [22 52]]
```

```
plt.figure(figsize=(6,5))
```

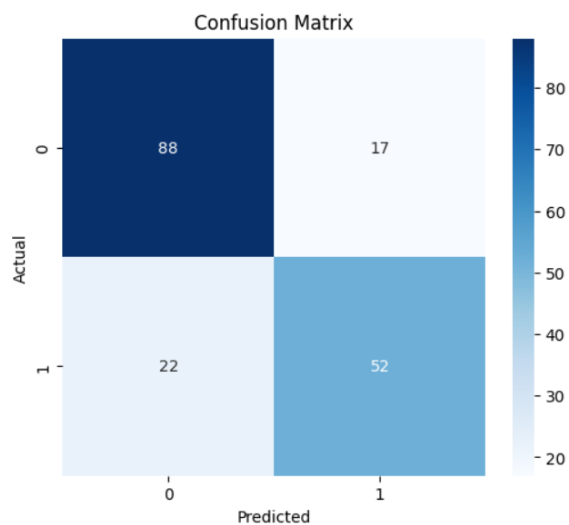
```
sns.heatmap(cm,  
            annot=True,  
            fmt='d',  
            cmap='Blues')
```

```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.title("Confusion Matrix")
```

```
plt.show()
```



Classification report :

Accuracy is the overall performance of the classification model and represents the percentage of correctly predicted instances out of the total test samples. In this model, the accuracy is 78%, meaning the model correctly classified 78% of the Titanic passengers.

Precision measures how many of the predicted instances for a class are actually correct. A higher precision indicates fewer false positive predictions. In this report, the precision is 0.80 for Class 0 and 0.75 for Class 1.

Recall measures how many actual instances of a class are correctly identified by the model. A higher recall indicates fewer false negatives. In this report, the recall is 0.84 for Class 0 and 0.70 for Class 1.

The F1-score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. It is 0.82 for Class 0 and 0.73 for Class 1, indicating better performance in predicting non-survivors.

Support represents the number of actual samples belonging to each class in the test dataset. In this report, Class 0 has 105 samples and Class 1 has 74 samples, making a total of 179 test instances.

Macro average is the unweighted average of precision, recall, and F1-score across all classes, treating each class equally. In this report, the macro average F1-score is 0.77.

Weighted average calculates the average precision, recall, and F1-score by considering the number of samples in each class. In this report, the weighted average values are 0.78, indicating good overall model performance.

- Discussion on which metric is most critical for medical diagnosis

Recall is the evaluation metric that is most commonly used for medical diagnosis. A high recall means that the disease in the patient is correctly detected. Misclassifying even a single patient can have a drastic negative effect in medical cases.

While other evaluation metrics are also important recall is the most critical in medical diagnosis. F1 score is also important.

- What accuracy measures

Accuracy measures the overall correctness of a classification model. It is the percentage of total predictions that are correct out of all predictions made. Accuracy is most useful when the dataset has a balanced distribution of classes.

- What precision measures and when it is important

Precision measures the proportion of positive predictions that are actually correct. It is important in situations where false positives are costly, such as spam email detection or fraud detection, where incorrectly classifying a negative case as positive can have undesirable consequences.

- What recall measures and when it is critical

To find the proportion of the actual positive cases that was identified by the algorithm we use recall. As discussed before it is given importance in medical cases where missing a patient can be can have extreme consequences

- How F1-score balances precision and recall

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of model performance by considering both false positives and false negatives. It is especially useful when both precision and recall are equally important or when dealing with imbalanced datasets.